

SHUBHAM SINGH

Senior Software Engineer

Engineer with 5 years of experience in Java/Spring Boot for scalable, high-availability, and reliable systems. Distributed Systems & Microservices, Reliability Engineering, and GenAI Enablement.

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Profile Summary

- Java/Spring Boot Backend Engineer with 5 years of experience designing, delivering, and scaling high-performance microservices and distributed systems.
- Hands-on experience with Spring Boot, Hibernate, JPA, and event-driven architectures. Skilled in Event-Driven Architectures (Kafka, SQS/SNS), Database Optimization (PostgreSQL, MongoDB, Redis), and building secure, multithreaded, and extensible enterprise solutions.
- Expertise in Reliability Engineering and Cloud-Native Architecture (AWS, Kubernetes, Docker), with demonstrated impact on production systems (e.g., reducing forceful pod terminations by 74%, zero-downtime MongoDB upgrades).
- Practical experience with AI/GenAI Enablement, preparing production systems for agentic workflows and LLM integration.

Technical Skills

- Production engineering: Observability, Incident skills, Performance tuning, Reliability patterns (bulkheads, rate limiters, load shedding, graceful degradation)
- Multithreading & Concurrency: Executors, ForkJoin, CompletableFuture
- Frameworks & Libraries: Spring Boot, Hibernate, JPA, Spring Security, Kafka, SQS/SNS, RabbitMQ
- Databases & Caching: MySQL, PostgreSQL, MongoDB, Redis, Redshift, Vector DB,, Memcached, Data modeling, indexing, query plans, partitions
- Cloud & DevOps: AWS, Docker, Kubernetes, CI/CD, Jenkins, Maven, Gradle, Shell Scripting, Git, Observability & Monitoring, Chaos Testing, Github Actions
- AI & Automation: AI/GenAI, Agentic Workflows, LLM integration, LangGraph, RAG, MCP standardization for agentic workflows, Codex, Claude Code, GitHub Copilot
- Testing: TDD, Mockito, JUnit
- Monitoring & Observability: DataDog, Splunk, Dynatrace, Grafana, ELK
- Microservices Internals: Config Server, API Gateway, Service Discovery, latency, timeouts, retries, backoff, circuit breakers
- Design Principles: SOLID, Design Patterns, Clean Architecture, HLD, LLD

Work Experience

Aug 2024 - Present | Expedia, Gurgaon, India | Software Engineer II

- Owned the Void/Cancel workflow end-to-end, designing idempotent APIs (idempotency keys + dedupe), retry-safe downstream calls (timeouts, exponential backoff, circuit breaker), and audit-grade tracking (state transitions + correlation IDs), enabling safe extensibility for new cancellation reasons/providers.
- Reduced Kubernetes forceful pod terminations by 74% (from 2.22k to 570) for a production backend service by correlating scale-down kills using Kubernetes events, validating shutdown behavior through chaos testing, and tuning.
- Led a high-risk MongoDB end-of-life upgrade for a tier-1 production service, migrating the cluster from v5 to v6 to v7 using rolling execution, ensuring zero downtime, and performing post-release validation through logs and performance dashboards.
- Worked on "Make Repositories AI-Ready" initiative, enabling GenAI and agentic workflows by standardizing code structure, improving metadata and documentation quality, and establishing foundations for scalable AI-driven development.

Nov 2023 - Aug 2024 | FIDEO, Gurgaon, India | Software Engineer

- Automated compliance processes for exporters and their customers by integrating third-party services, improving operational efficiency, and integrating the complete flow with OpenAI APIs.
- Mentored 4 junior developers on integration architecture and distributed systems patterns, resulting in a 50% increase in team delivery velocity and improved code quality.
- Built Customer Aging Reporting system processing \$50M+ in daily transactions, reducing reporting latency by 40% via a new API-digestible architecture.

Sep 2021 - Aug 2023 | PAYU, Gurgaon, India | Software Engineer

- Implemented a maker-checker workflow for the Pricing Engine (Pricing-as-a-Service), including bulk processing and fail-safe mechanisms to ensure process reliability.
- Streamlined settlement summary and transaction download processes, improving transparency and efficiency for merchants and clients.

Education & Credentials

- B.Tech. in Computer Science|
September 2021 | Noida, India

Active Personal Project

RAG Notebook | AI-Native Distributed Knowledge Base (In Development / V 1.0)

Tech Stack: Java 21 (Spring Boot), Python (LangGraph), PostgreSQL (pgvector), Docker, Redis.

- Architecting a polyglot microservices system where a **Spring Boot API handles high-concurrency user requests and offloads heavy document processing to Python workers via an async event queue (Redis/Queue).**
- Building a "Model Gateway" to abstract LLM providers, allowing the system to hot-swap between local models (Ollama/Phi-3) for zero-cost development and frontier models (Claude/GPT-4o) for production quality.
- Implementing confidence-based routing to optimize inference costs; **currently tuning thresholds where the system attempts retrieval with budget models first and only escalates to expensive models when semantic similarity scores are low.**
- Designing the "**Groundedness Loop**" to catch hallucinations; currently instrumenting an async critic that grades every answer against retrieved chunks to drive automated index improvements.

Soft Skills

Leadership Competence,
Interpersonal Skills, Effective
Decision-Making, Analytical,
Problem-Solving,
Conflict Resolution Expertise